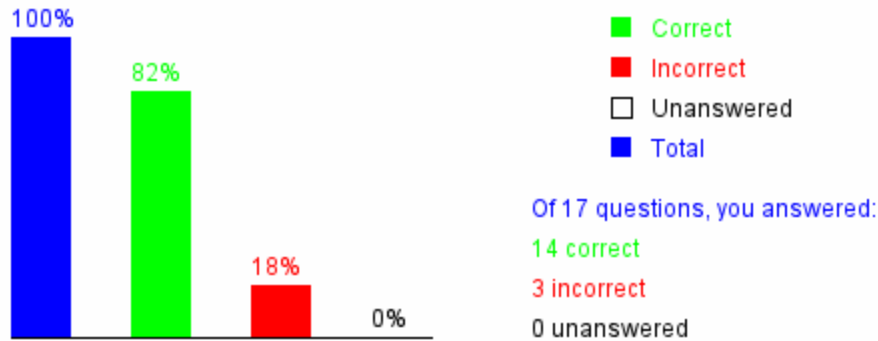


Introduction to Programming Using Python, Y. Daniel Liang

Many questions in this edition have been updated in the new edition. Please check with the publisher on the newest edition.

This quiz is for students to practice. A large number of additional quiz is available for instructors using Quiz Generator from the Instructor's Resource Website. Videos for Java, Python, and C++ can be found at <https://yongdanielliang.github.io/revelvideos.html>.

Chapter 11 Multidimensional Lists



Please send suggestions and errata to Dr. Liang at y.daniel.liang@gmail.com. Indicate which book and edition you are using.
 Thanks!

11.1 What will be displayed by the following code?

```
m = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
print(m[0][0])
```

- ☒ A. 1
- ☐ B. 2
- ☐ C. 4
- ☐ D. 7

Your answer is correct



11.2 Assume `m = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]`, what are `len(m)`?

- ☐ A. 0
- ☐ B. 1
- ☐ C. 2
- ☒ D. 3
- ☐ E. 4

Your answer is correct



11.3 Assume `m = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]`, what are `len(m[0])`?

- ☐ A. 0
- ☐ B. 1
- ☐ C. 2
- ☒ D. 3
- ☐ E. 4

Your answer is correct



11.4 For `m = [[x, x + 1, x + 2] for x in range(0, 3)]`, `m` is _____.

- ☐ A. `[[1, 2, 3], [4, 5, 6], [7, 8, 9]]`
☒ B. `[[0, 1, 2], [1, 2, 3], [2, 3, 4]]`
☐ C. `[1, 2, 3, 4, 5, 6, 7, 8, 9]`
☐ D. `[0, 1, 2, 1, 2, 3, 2, 3, 4]`

Your answer is correct



11.5 For `m = [[x, x + 1, x + 2] for x in range(1, 9, 3)]`, `m` is _____.

- ☐ A. `[[1, 2, 3], [4, 5, 6], [7, 8, 9]]`
☒ B. `[[0, 1, 2], [1, 2, 3], [2, 3, 4]]`
☐ C. `[1, 2, 3, 4, 5, 6, 7, 8, 9]`
☐ D. `[0, 1, 2, 1, 2, 3, 2, 3, 4]`

Your answer B is incorrect



[Click here to show the correct answer](#)

11.6 How many elements are in `m = [[x, y] for x in range(0, 4) for y in range(0, 4)]`?

- ☐ A. 8
☐ B. 12
☒ C. 16
☐ D. 32

Your answer is correct



11.7 Assume `x = ((1, 2), (3, 4, 5), (5, 6, 5, 9))`, what are `len(x)` and `len(x[0])`?

- ☐ A. 2 and 1
☐ B. 2 and 2
☒ C. 3 and 2
☐ D. 2 and 3
☐ E. 3 and 3

Your answer is correct



11.8 Assume `x = [[1, 2], [3, 4, 5], [5, 6, 5, 9]]`, what are `len(x[0])`, `len(x[1])`, and `len(x[2])`?

- ☐ A. 2, 3, and 3
☒ B. 2, 3, and 4
☐ C. 3, 3, and 3
☐ D. 3, 3, and 4
☐ E. 2, 2, and 2

Your answer is correct



11.9 What will be displayed by the following program?

```

values = [[3, 4, 5, 1], [33, 6, 1, 2]]

v = values[0][0]
for row in range(0, len(values)):
    for column in range(0, len(values[row])):
        if v < values[row][column]:
            v = values[row][column]
  
```

```
print(v)
```

- ☐ A. 1
- ☐ B. 3
- ☐ C. 5
- ☐ D. 6
- ☒ E. 33

Your answer is correct



11.10 What will be displayed by the following program?

```
values = [[3, 4, 5, 1], [33, 6, 1, 2]]
```

```
v = values[0][0]
for lst in values:
    for element in lst:
        if v > element:
            v = element
```

```
print(v)
```

- ☐ A. 1
- ☐ B. 3
- ☐ C. 5
- ☐ D. 6
- ☒ E. 33

Your answer E is incorrect



[Click here to show the correct answer](#)

11.11 What will be displayed by the following program?

```
values = [[3, 4, 5, 1 ], [33, 6, 1, 2]]
```

```
for row in values:
    row.sort()
    for element in row:
        print(element, end = " ")
    print()
```

- ☐ A. The program prints two rows 3 4 5 1 followed by 33 6 1 2
- ☐ B. The program prints on row 3 4 5 1 33 6 1 2
- ☐ C. The program prints two rows 3 4 5 1 followed by 33 6 1 2
- ☒ D. The program prints two rows 1 3 4 5 followed by 1 2 6 33
- ☐ E. The program prints one row 1 3 4 5 1 2 6 33

Your answer is correct



11.12 What will be displayed by the following code?

```
matrix = [[1, 2, 3, 4],
          [4, 5, 6, 7],
          [8, 9, 10, 11],
          [12, 13, 14, 15]]
```

```
for i in range(0, 4):
    print(matrix[i][1], end = " ")
```

- ☐ A. 1 2 3 4
- ☐ B. 4 5 6 7
- ☐ C. 1 3 8 12
- ☒ D. 2 5 9 13
- ☐ E. 3 6 10 14

Your answer is correct



11.13 What will be displayed by the following code?

```
matrix = [[1, 2, 3, 4],
          [4, 5, 6, 7],
          [8, 9, 10, 11],
          [12, 13, 14, 15]]

for i in range(0, 4):
    print(matrix[1][i], end = " ")
```

- ☐ A. 1 2 3 4
- ☒ B. 4 5 6 7
- ☐ C. 1 3 8 12
- ☐ D. 2 5 9 13
- ☐ E. 3 6 10 14

Your answer is correct



11.14 What will be displayed by the following program?

```
def m(list):
    v = list[0]
    for e in list:
        if v < e: v = e
    return v

values = [[3, 4, 5, 1], [33, 6, 1, 2]]

for row in values:
    print(m(row), end = " ")
```

- ☒ A. 3 33
- ☐ B. 1 1
- ☐ C. 5 6
- ☐ D. 5 33
- ☐ E. 33 5

Your answer A is incorrect



[Click here to show the correct answer](#)

11.15 What will be displayed by the following code?

```
data = [[[1, 2], [3, 4]], [[5, 6], [7, 8]]]

print(data[1][0][0])
```

- ☐ A. 1
- ☐ B. 2
- ☐ C. 4

- ☒ D. 5
☐ E. 6

Your answer is correct



11.16 What will be displayed the following code?

```
data = [[1, 2], [3, 4]], [[5, 6], [7, 8]]

def ttt(m):
    v = m[0][0]

    for row in m:
        for element in row:
            if v < element: v = element

    return v

print(ttt(data[0]))
```

- ☐ A. 1
☐ B. 2
☒ C. 4
☐ D. 5
☐ E. 6

Your answer is correct



11.17 What will be displayed by the following code?

```
points = [[1, 2], [3, 1.5], [0.5, 0.5]]
points.sort()
print(points)
```

- ☐ A. [[1, 2], [3, 1.5], [0.5, 0.5]]
☐ B. [[3, 1.5], [1, 2], [0.5, 0.5]]
☒ C. [[0.5, 0.5], [1, 2], [3, 1.5]]
☐ D. [[0.5, 0.5], [3, 1.5], [1, 2]]

Your answer is correct

